

Hypothesis Testing And Statistical Analysis

- ① Z Test } $\Rightarrow$ Average $\Rightarrow$ Z table $\rightarrow$ Z score And p value
- ② t Test } $\Rightarrow$ t table
- ③ CHI SQUARE $\Rightarrow$ Categorical Data
- ④ ANNOVA $\Rightarrow$ Variance

We can apply z-test only & only if this 2 condition states

Z test . i) Population std ii) $n \geq 30$

With a $\sigma = 3.9$

i) The average heights of all residents in a city is 168cm. A doctor believes the mean to be different. He measured the height of 36 individuals and found the average height to be 169.5 cm.

(a) State null and Alternate Hypothesis

(b) At a 95% confidence level, is there enough evidence to reject the null hypothesis.

Ans) $\mu = 168\text{cm}$ $\sigma = 3.9$ $n = 36$ $\bar{x} = 169.5$

C.I = 0.95
Confidence Interval

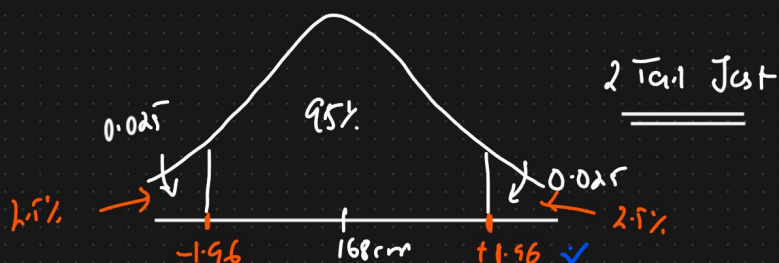
$\alpha = 1 - C.I = 1 - 0.95 = 0.05$
Significance value

alpha = P value

① Null Hypothesis $H_0 = \mu = 168\text{cm}$

② Alternate Hypothesis $H_1 = \mu \neq 168\text{cm}$

③ Based on C.I we will draw Decision Boundary



$$1 - 0.025 = 0.9750 \Rightarrow Z\text{-score}$$

Z - score for right one

↓

$$\text{Area} \Rightarrow +1.96$$

Z - score derived from z table

Goto: <https://www.ztable.net/> -> See 1.9 & 0.6

if Z is less than -1.96 or greater than $+1.96$, Reject the Null Hypothesis.

⊗ Z-test

$$Z_d = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$$Z_{\text{score}} = \frac{\bar{X} - \mu}{\sigma}$$

↓

$$\frac{\sigma}{\sqrt{n}}$$

Sample S.D.

$$= \frac{169.5 - 168}{3.9 / \sqrt{36}}$$

$$Z_d = \frac{1.5}{0.65} = 2.31$$

Conclusion

$$2.31 > 1.96$$

Reject the Null Hypothesis

The P - value is = Significance value (alpha)

$$p < 0.05$$

Now we see inversly into the z-table

Watching 2.3 & 0.1 after derived the value of 0.98956



$$0.98956$$

Now to get remaining area

$$1 - 0.98956 = 0.01044$$

⊗ Final Conclusion the Average $\neq$ 168cm

The average height seems to increasing based on sample height.

$$\textcircled{1} \text{ p value} = 0.01044 + 0.01044$$

$$= 0.02088$$

$$p < 0.05$$

$0.02088 < 0.05 \Rightarrow$ Reject the Null Hypothesis

② A factory manufactures bulbs with a average warranty of 5 years with standard deviation of 0.50. A worker believes that the bulb will malfunction in less than 5 years. He tests a sample of 40 bulbs and find the average time to be 4.8 years.

(a) State null and alternate hypothesis

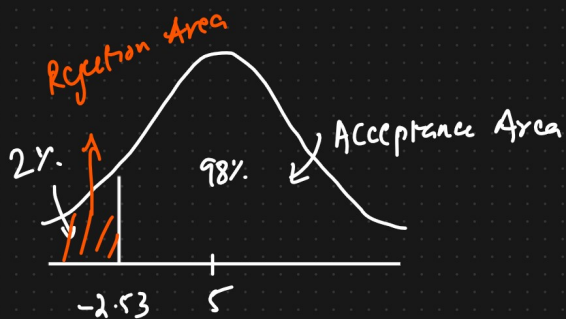
(b) At a 2% significance level, is there enough evidence to support the idea that the warranty should be revised?

Ans) $\mu = 5$ $\sigma = 0.50$ $n = 40$ $\bar{x} = 4.8$
 0.5

a) Null hypothesis $H_0: \mu = 5$

Alternate Hypothesis $H_1: \mu < 5$ {1 Tail Test}

5) Decision Boundary



c) Z-test

$$Z_d = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{4.8 - 5}{0.50 / \sqrt{40}} = -2.53$$

Area under curve with Z score $-2.53 = 0.0570$.

$$P\text{-value} = 0.0570$$

$$\alpha = 0.02$$

Compare P-value with α

$0.0570 < 0.02 \Rightarrow$ False

We accept the Null Hypothesis

We Fail to Reject the Null Hypothesis.